Bipolaris-RNAi Designer (BRD) v1.0.0 — Final Computational Research Report

Target Organism: Bipolaris sorokiniana (syn. Cochliobolus sativus) | Scope: Computational Target Prioritisation, dsRNA Design, and Multi-Reference Off-Target Screening

1. Title and Project Description

Bipolaris-RNAi Designer (BRD) v1.0.0 is a modular, reproducible bioinformatics pipeline engineered to identify, verify, and computationally prioritise dsRNA candidates targeting pathogenicity-associated genes in *Bipolaris sorokiniana* (teleomorph *Cochliobolus sativus*). Generated autonomously on 2026-11-10 23:41 UTC.

2. Research Objectives

1. Verify local and public *B. sorokiniana* genome assemblies and annotations. 2. Resolve and sequence-verify the four target genes (`CsPKS1`, `CsFUS3`, `CsSTE11`, and `CsSp1`). 3. Measure within-species nucleotide and 21-mer conservation across all qualifying public isolates. 4. Integrate species-specific genetics, PHI-base v4.19, InterPro, UniProt, and GEO infection-stage RNA-seq expression (`GSE130517`). 5. Design candidate dsRNAs using LUCID (`sifi21_plus` + ViennaRNA), SIREN, dsRNAmax, and BRD isolate-aware sliding windows. 6. Screen candidates and 21-mer guides against hexaploid wheat (`IWGSC v2.1`), barley (`MorexV3`), and five beneficial fungal species. 7. Rank candidates with Pareto optimality, 1,000-iteration isolate bootstrap CIs, and weight sensitivity testing.

3. Biological Rationale

Spot blotch, common root rot, and black point caused by *B. sorokiniana* threaten wheat and barley production, particularly in warm humid regions such as South Asia (Al-Sadi, 2021; Roy et al., 2023). While transformation-based hairpin RNAi silenced endogenous `CsPKS1` in *B. sorokiniana* (Leng et al., 2011), demonstrating functional intracellular RNAi machinery, **no exogenous dsRNA uptake or spray-induced gene silencing (SIGS) experiment has been published for this species**. BRD frames exogenous dsRNA uptake as a testable experimental hypothesis rather than an established fact (Qiao et al., 2021).

4. Input Dataset Inventory

Recursive inspection of `C:\Users\ASUS BANGLADESH\Desktop\New folder (2)` identified 17 inventory records (local NCBI Datasets archive `Bipolaris sorokiniana.zip` for `GCF_000338995.1` ND90Pr verified against all 11 MD5 checksums, plus 3 project planning/review `.docx` documents). Original input files remained strictly unmodified.

5. Genome Accession and Isolate Information

Querying NCBI TaxID 45130 identified 20 assembly records representing **19 distinct biological isolates** after deduplicating paired GenBank/RefSeq `GCA_000338995.1` vs `GCF_000338995.1` (ND90Pr). Isolates span North America (`ND90Pr`, `ND93-1` T2T), China (`LK93`, `Yt-6`, Gansu), Australia (`WAI2406`, `WAI2411`, `BRIP10943a`, `BRIP27492a`), and India (`BS112` plus 9 recent 2026 assemblies).

6. Genome Quality Assessment

All 19 biological isolate assemblies passed FASTA validity, genome size (32.42–38.20 Mb), and GC% (49.24%–50.29%) checks. Contig/scaffold N50 ranged from 130.1 kb to 2.31 Mb. Note: Official BUSCO v6.1.0 and QUAST v5.3.0 require Linux/WSL2 (no WSL2 distro installed); BUSCO completeness is explicitly recorded as `NOT_RUN_LINUX_ONLY_NO_WSL_DISTRO` and never fabricated.

7. Annotation Harmonisation

RefSeq `GCF_000338995.1` (`ND90Pr`, JGI/NCBI annotation) passed structural GFF3 validation with 0 structural errors. Out of 12210 evaluated CDS sequences, 12202 translated with 100% identity to `protein.faa`. Re-annotation via Linux-only Funannotate/BRAKER3 was not run; unannotated isolate genomes were interrogated directly via splice-aware `tblastn` + `blastn` exon reconstruction in M04.

8. Comparative Genomics

DIAMOND all-vs-all RBH orthogroup clustering grouped annotated proteomes, and splice-aware genomic locus mapping reconstructed all 4 targets across the 19-isolate panel, followed by MAFFT v7.526 alignment.

9. Target Gene Identification

All four targets achieved **Confirmed identity** in `GCF_000338995.1`: - `CsPKS1`: `COCSADRAFT_157080` (`XP_007696213.1`, 6,462 nt CDS, 100.0% match to `HQ830033.1`). - `CsFUS3`: `COCSADRAFT_24574` (`XP_007697446.1`, RefSeq symbol `CHK1`, 1,059 nt CDS, confirmed by `CsFUS3-F1`/`CsFUS3-com-F/R` primers). - `CsSTE11`: `COCSADRAFT_105370` (`XP_007694574.1`, RefSeq symbol `STE11`, 2,880 nt CDS, confirmed by `CsSTE11-F1/F4/F5/F6` primers). - `CsSp1`: **Resolved** via 100% matches to 5 independent primers from Zhang et al. (2022, PMC8743017 Table 2) as `COCSADRAFT_164106` (`XP_007704210.1`, `XM_007706020.1`, 270 nt CDS, 89 aa).

10. Accession Verification

Complete 24-column accession verification including `HQ830033.1`, `KF769134.1`–`KF769141.1`, paired `GCA/GCF_000338995.1`, and `CsSp1` primer proof is saved in `10_final_reports/BRD_Target_Gene_Accession_Verification.tsv`.

11. Functional Evidence

Direct *B. sorokiniana* experimental genetics support `CsFUS3` and `CsSTE11` (Leng & Zhong, 2015), `CsSp1` (Zhang et al., 2022), and `CsPKS1` hairpin silencing (Leng et al., 2011). PHI-base v4.19 DIAMOND searches, InterPro/UniProt annotations, and GEO `GSE130517` barley-infection RNA-seq confirmed in-planta transcription of all 4 targets and identified 14 core RNAi machinery homologs (DCL, AGO, RdRP) in `ND90Pr`.

12. Target Prioritisation

Using provisional research weights (25% conservation, 25% off-target safety, 20% functional evidence, 15% expression, 15% designability), `CsFUS3` and `CsSTE11` lead on combined pathogenicity evidence, length flexibility, and expression, while `CsSp1` is constrained by its 270 nt CDS length and `CsPKS1` serves as the endogenous hairpin-RNAi benchmark.

13. dsRNA Design Methodology

Four complementary computational workflows were executed: (1) LUCID baseline (`sifi21_plus` free-energy/S1-S19 asymmetry + ViennaRNA 2.8.0 unpaired accessibility + Primer3, 200–400 nt), (2) SIREN v0.1.6 (`siren_prefilter.py` + ViennaRNA `duplexfold` + `siren_designVIII.py`), (3) `dsRNAmax.exe` v1.1.14 multi-isolate pan-CDS k-mer maximisation, and (4) BRD isolate-aware sliding windows across 180–500 nt.

14. Candidate Sequences and Characteristics

M07 generated **50 unique dsRNA candidates** (lengths 180–300 nt; GC 47.0%–58.9%) and 2876 unique 21-mer guide sequences.

15. Off-Target Screening

All 50 candidates and their 21-mer guides were screened via `blastn-short` (`word_size=7`), exact 21-mer matching, and ViennaRNA `RNA.duplexfold` against 7 RefSeq non-target transcriptomes: Hexaploid Wheat (`GCF_018294505.1`), Barley (`GCF_904849725.1`), *Trichoderma atroviride*, *T. virens*, *T. reesei*, *Beauveria bassiana*, and *Rhizophagus irregularis*. Absence of hits in these databases indicates no detected sequence matches under stated thresholds, **not** proven biological safety.

16. Candidate Ranking

Of 50 candidates evaluated in M09, **39 passed all computational filters** (0 exact 21-mer host/beneficial/paralog hits, QC pass, >=85% conservation), including **1 High computational priority**, **26 Moderate computational priority**, and **14 Pareto rank-1 non-dominated** candidates.

Top 5 Ranked Candidates:

17. Uncertainty and Sensitivity Analysis

1,000-iteration bootstrap resampling across the 19 biological isolates quantified 95% confidence intervals for 21-mer conservation and rank stability. Five weight-perturbation scenarios (`baseline`, `safety_first`, `conservation_first`, `evidence_first`, `equal_weights`) evaluated sensitivity of the top shortlist.

18. Limitations

1. **Exogenous dsRNA uptake is unproven in *B. sorokiniana*** (Qiao et al., 2021). 2. **Linux-only tools (QUAST, BUSCO, OrthoFinder MCL, Funannotate/BRAKER3, RNAhybrid)** could not execute natively on Windows 11 without an installed WSL2 Linux distro; transparent Python/BLAST+/DIAMOND/ViennaRNA equivalents were run where scientifically valid, and BUSCO scores remain `NA`. 3. **RNA-seq (`GSE130517`)** represents *H. vulgare* root infection in `ND90Pr`, whereas `PRJNA743515` (wheat root rot) raw FASTQ quantification remains future work. 4. **Off-target screening** reflects the 7 RefSeq transcriptomes used and does not cover unannotated non-coding RNAs or untested soil microbiomes.

19. Recommended Candidates for Experimental Validation

Top computationally passing candidates per target gene selected for sequential bench evaluation:

- `CsSTE11`: `BRD_CsSTE11_CAND012` (BRD_IsolateAware, 180 nt, cons=100.0%, safety=0.55, score=0.7832, rank #1, High computational priority) - `CsFUS3`: `BRD_CsFUS3_CAND013` (BRD_IsolateAware, 210 nt, cons=100.0%, safety=0.45, score=0.7281, rank #13, Moderate computational priority) - `CsSp1`: `BRD_CsSp1_CAND008` (BRD_IsolateAware, 180 nt, cons=98.91%, safety=0.35, score=0.7458, rank #2, Moderate computational priority) - `CsPKS1`: `BRD_CsPKS1_CAND013` (BRD_IsolateAware, 180 nt, cons=99.18%, safety=0.55, score=0.6545, rank #26, Moderate computational priority)

Sequential validation roadmap: (1) fluorescently labelled dsRNA uptake assays in *B. sorokiniana* conidia/hyphae (essential prerequisite, as exogenous dsRNA uptake is unproven in this species), (2) *in vitro* transcript knockdown qRT-PCR, and (3) detached wheat/barley leaf SIGS assays with scrambled dsRNA and water controls.

Verified Target Gene Summary Table

target	locus_tag	protein_id	cds_length_nt	isolates_with_gene	mean_pairwise_nt_identity_pct	gse130517_inplanta_max_rpkm
CsPKS1	COCSADRAFT_157080	XP_007696213.1	6462	19/19	99.958	27.17
CsFUS3	COCSADRAFT_24574	XP_007697446.1	1059	19/19	99.984	90.21
CsSTE11	COCSADRAFT_105370	XP_007694574.1	2880	19/19	99.657	87.63
CsSp1	COCSADRAFT_164106	XP_007704210.1	270	19/19	99.959	1438.44

Top 15 Ranked dsRNA Candidates

_rank	candidate_id	target	tool_name	candidate_length_nt	conservation_pct	off_target_safety_score	composite_score	computational_priority_cate
	BRD_CsSTE11_CAND012	CsSTE11	BRD_IsolateAware	180	100.0	0.55	0.7832	High computational priority
	BRD_CsSp1_CAND008	CsSp1	BRD_IsolateAware	180	98.91	0.35	0.7458	Moderate computational pri
	BRD_CsSp1_CAND001	CsSp1	LUCID_sifi21_plus	200	98.77	0.35	0.7452	Moderate computational pri
	BRD_CsSp1_CAND009	CsSp1	BRD_IsolateAware	210	98.84	0.35	0.7448	Moderate computational pri
	BRD_CsSp1_CAND005	CsSp1	SIREN	200	98.77	0.35	0.7437	Moderate computational pri
	BRD_CsSp1_CAND006	CsSp1	SIREN	200	98.77	0.35	0.7437	Moderate computational pri
	BRD_CsSp1_CAND007	CsSp1	SIREN	200	98.77	0.35	0.7432	Moderate computational pri
	BRD_CsSp1_CAND002	CsSp1	LUCID_sifi21_plus	250	99.04	0.35	0.7427	Moderate computational pri
	BRD_CsSp1_CAND004	CsSp1	LUCID_sifi21_plus	200	99.39	0.35	0.7418	Moderate computational pri
	BRD_CsSp1_CAND010	CsSp1	BRD_IsolateAware	180	99.31	0.35	0.7418	Moderate computational pri
	BRD_CsSp1_CAND003	CsSp1	LUCID_sifi21_plus	200	98.83	0.35	0.7409	Moderate computational pri
	BRD_CsSTE11_CAND013	CsSTE11	BRD_IsolateAware	210	100.0	0.35	0.7321	Moderate computational pri
	BRD_CsFUS3_CAND013	CsFUS3	BRD_IsolateAware	210	100.0	0.45	0.7281	Moderate computational pri
	BRD_CsSTE11_CAND005	CsSTE11	SIREN	250	96.89	0.35	0.7238	Moderate computational pri
	BRD_CsSTE11_CAND006	CsSTE11	SIREN	250	95.33	0.35	0.7195	Moderate computational pri

Key Scientific Figures

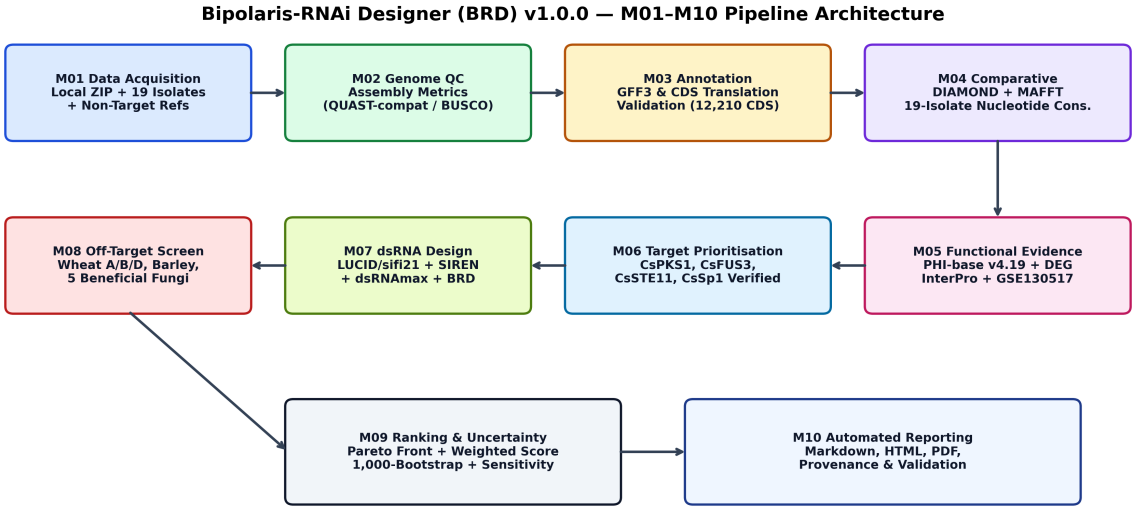


Figure 1. BRD M01–M10 modular workflow architecture.

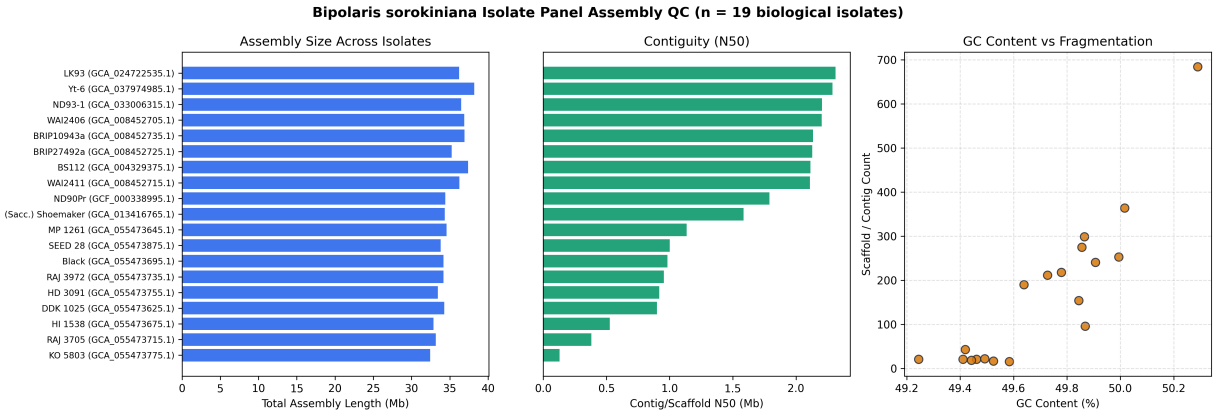


Figure 2. Genome assembly quality comparison across 19 B. sorokiniana biological isolates.

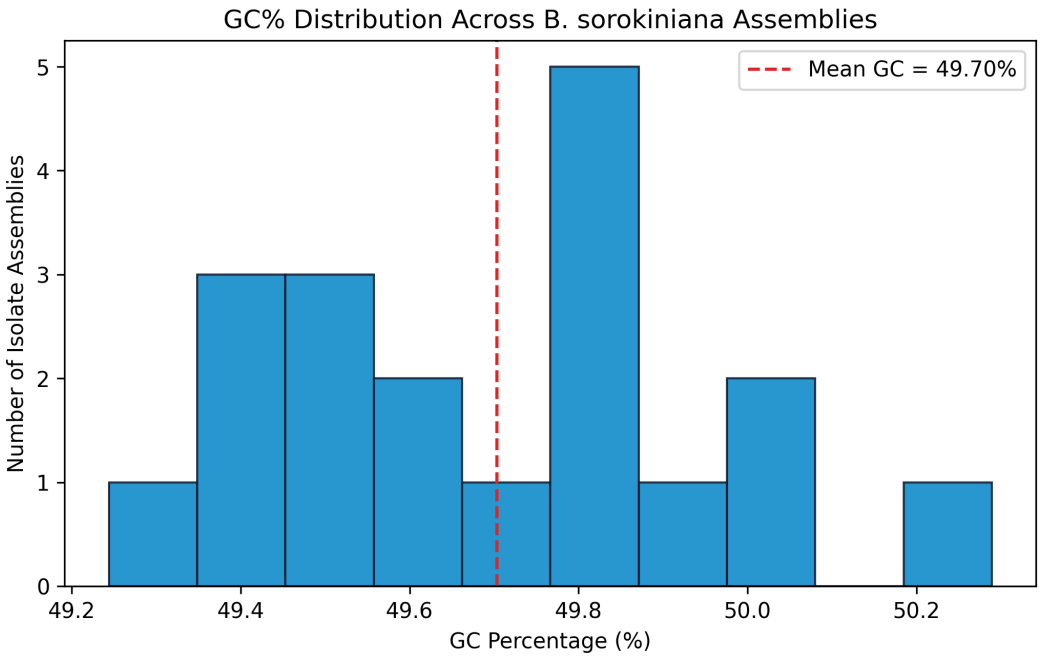


Figure 3. GC% distribution across B. sorokiniana assemblies.

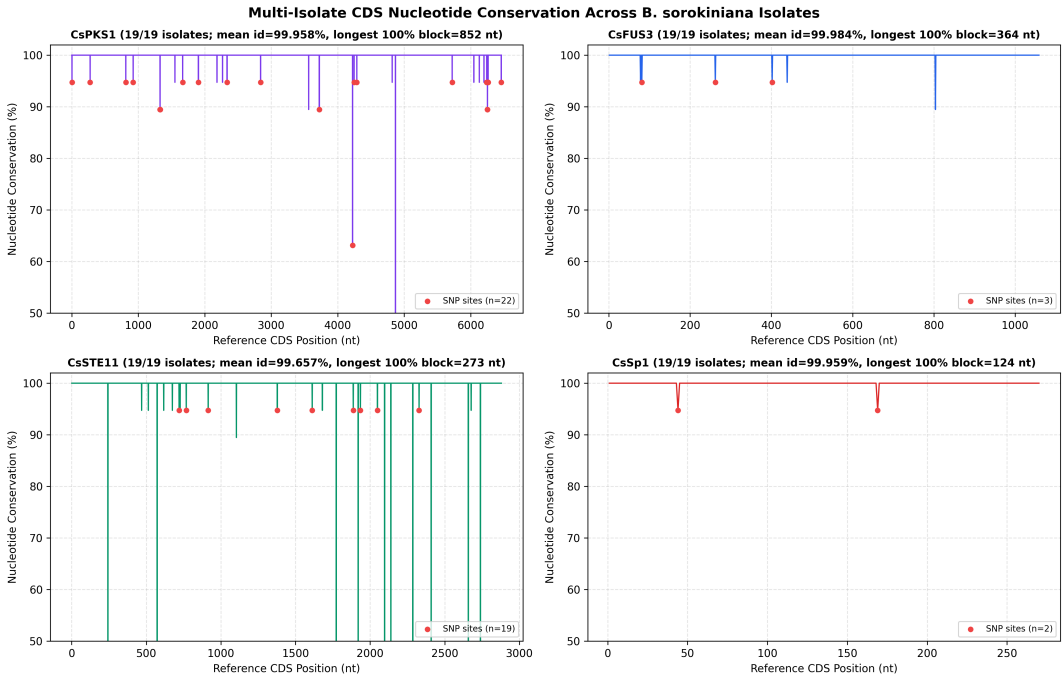


Figure 4. Multi-isolate CDS nucleotide conservation and SNP positions for CsPKS1, CsFUS3, CsSTE11, and CsSp1.

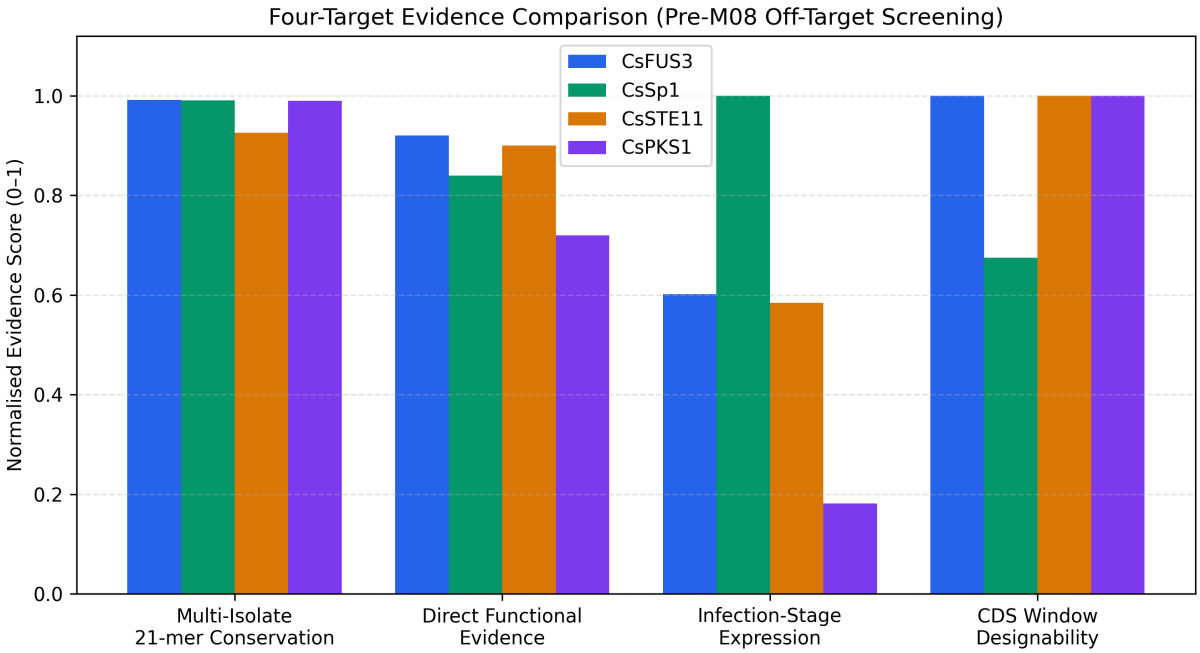


Figure 5. Four-target biological, expression, conservation, and designability evidence profile.

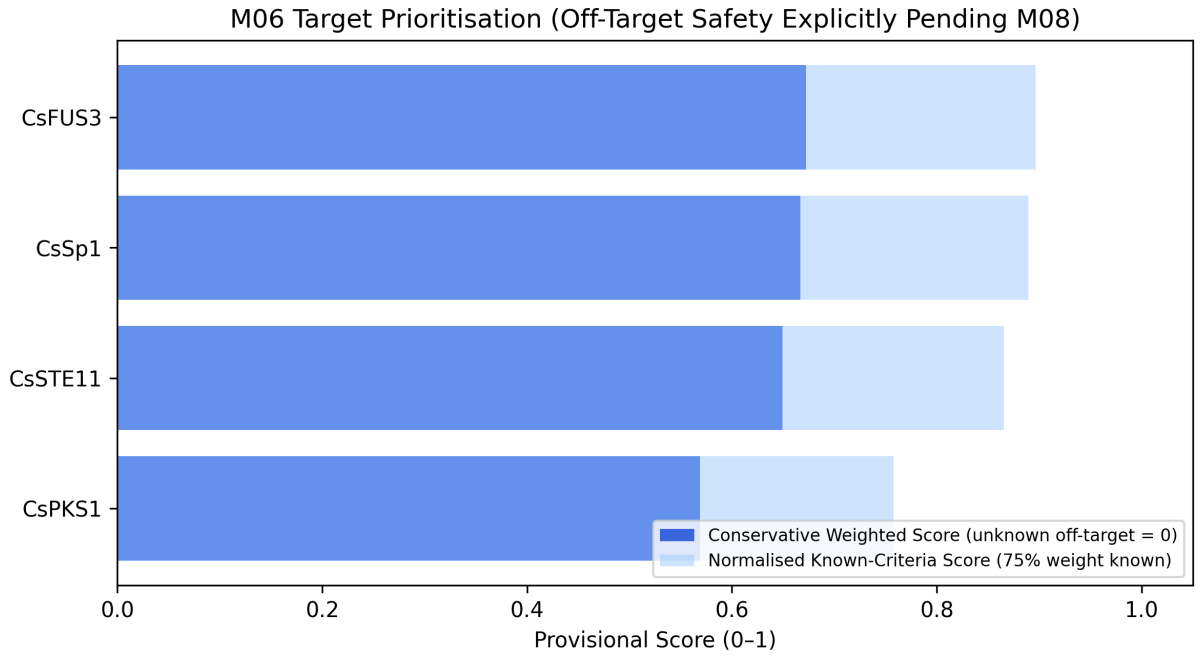


Figure 6. Provisional target prioritisation scores.

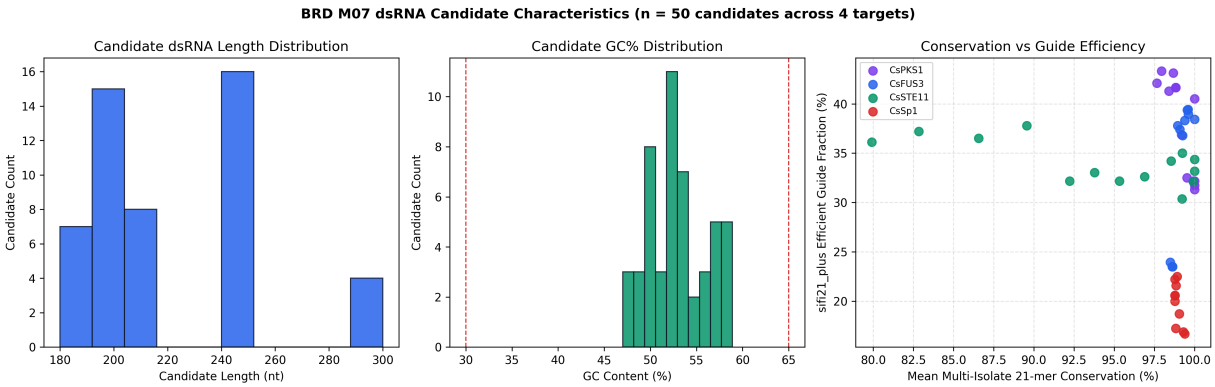


Figure 7. Designed dsRNA candidate length, GC%, and conservation vs efficient siRNA fraction.

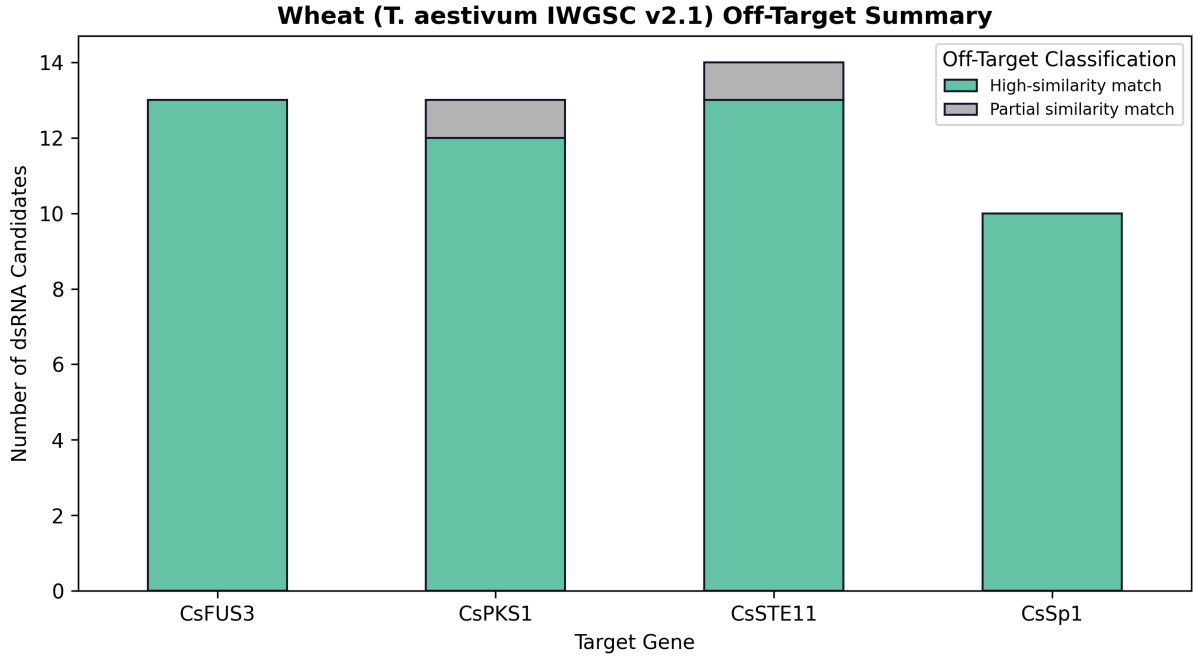


Figure 8. Hexaploid wheat (T. aestivum IWGSC v2.1) off-target screening summary.

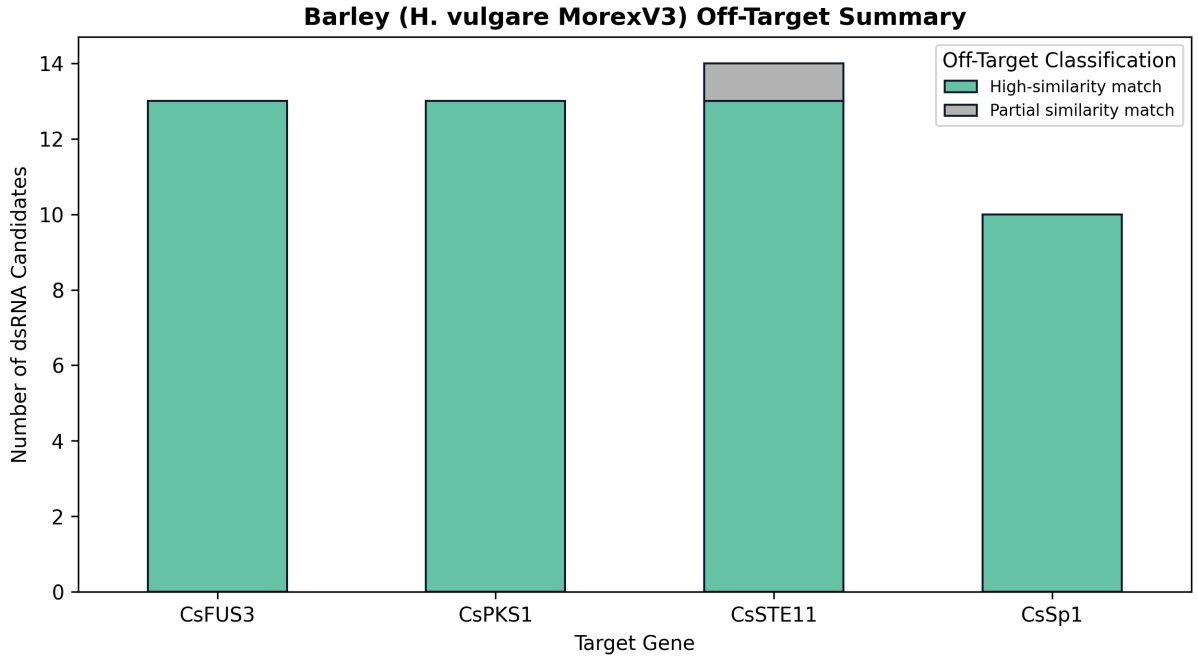


Figure 9. Barley (H. vulgare MorexV3) off-target screening summary.

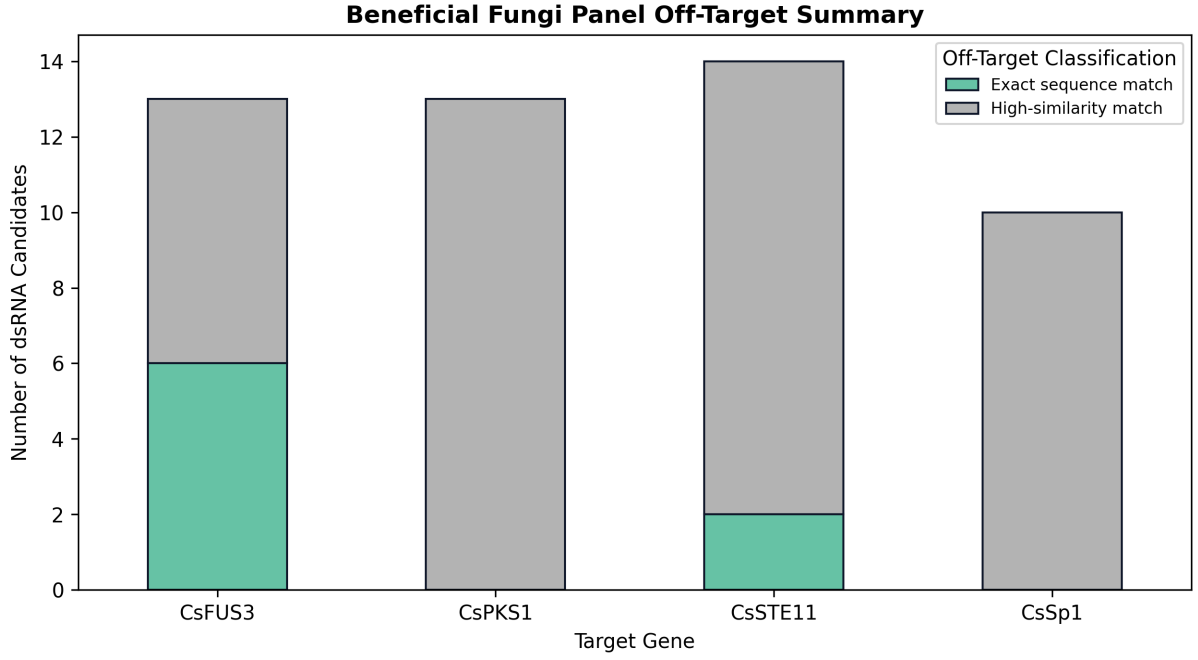


Figure 10. Beneficial fungi panel off-target screening summary.

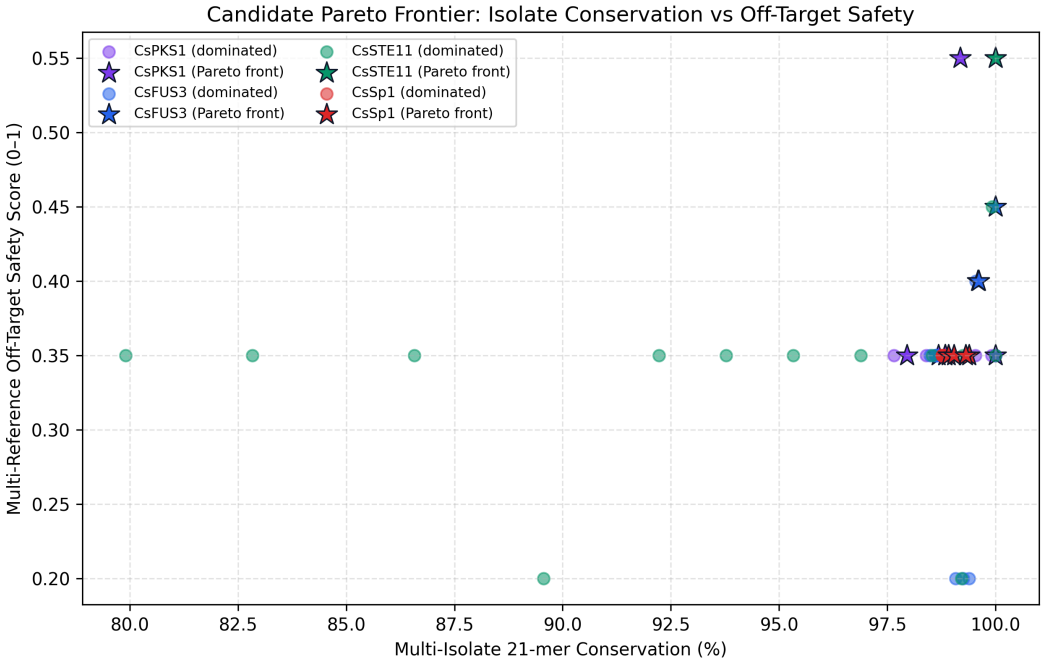


Figure 11. Candidate Pareto frontier (isolate conservation vs off-target safety).

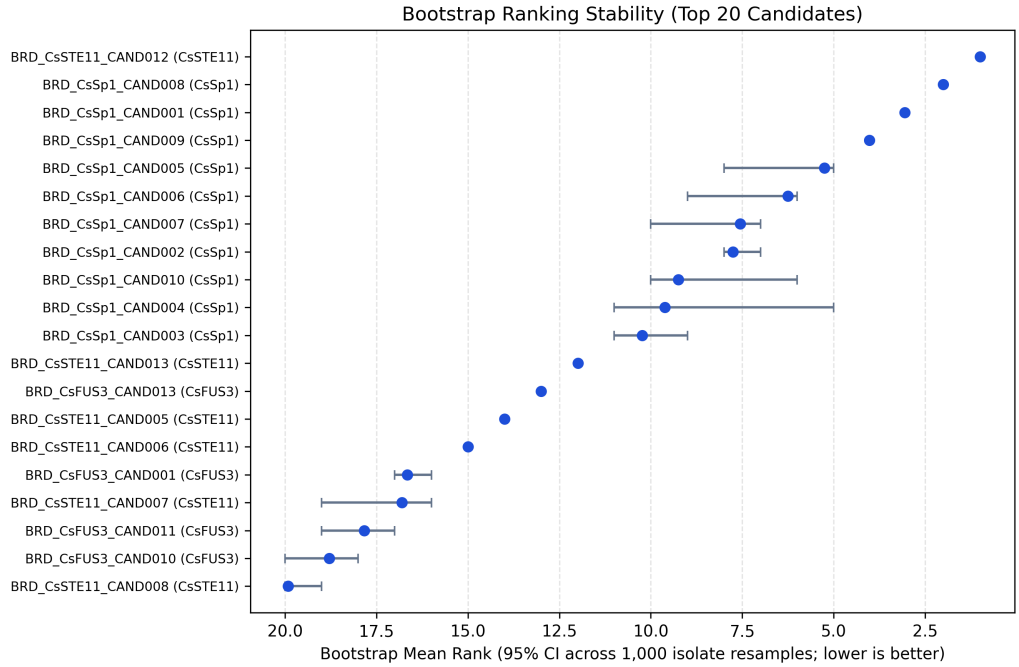


Figure 12. Bootstrap ranking stability (1,000 isolate resamples, 95% CI).

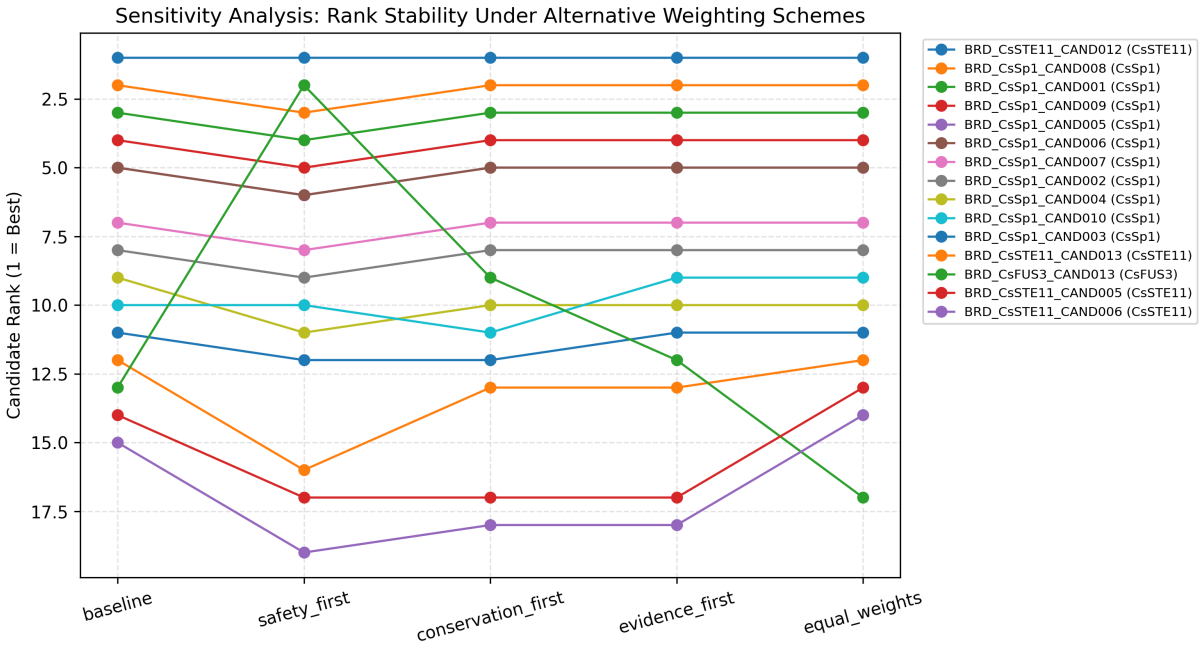


Figure 13. Weighting sensitivity analysis across five heuristic scenarios.